Research Question and Approach

Gabriel Gomez Hernandez

Research Question

Energy consumption can vary significantly across households, due to factors such as the type of appliances, their energy efficiency, and the primary source of fuel for these appliances that are used throughout the day. With this in mind, it is particularly interesting to explore the patterns that are generated from the different peaks in energy that arise at various times of the day.

Observations from the initial exploratory analysis suggest that these patterns can be related to household routines, appliance use, and lifestyle factors. However, something remained unclear from the initial observations, the extent to which energy-use profiles exist. Preliminary plots and summary statistics indicate that there are identifiable periods in time where usage could be grouped based on the peaks to reveal meaningful patterns. Thus, to explore this gap in the research and further support previous observations, the following research question arises: *Can households be grouped based on their daily peak electricity and gas usage hours to identify distinct energy-use profiles?*

Approach

To answer the research question, I plan to use the k-means cluster algorithm as it is an unsupervised and global technique that finds meaningful groups which can be used to group households based on their daily peak electricity and gas usage hours. Clustering is appropriate for this analysis as it allows us to identify natural groupings without needing predefined labels and as it was mentioned previously, this aligns with the goal of discovering meaningful and distinct energy-use profiles.

Key variables:

- Hourly electricity consumption
- Hourly gas consumption
- Date-time (days and hours)

These variables and chosen method can help identify similarities in households' daily consumption patterns and based on the resulting groups, establish distinct energy-use profiles. These profiles can then be used to develop strategies for energy efficiency improvements and better management of energy demands.